

Star-Finite Coverings and the Star-Finite Property.

By Kiiti MORITA.

A covering $\mathcal{U}$ of a T_1 -space is called to be star-finite, if each set of $\mathcal{U}$ intersects a finite number of other sets of $\mathcal{U}$. In case every open covering of a T_1 -space R admits a star-finite open refinement we shall say that R possesses the star-finite property. Then it will be shown that for a regular space R the star-finite property is implied by the Lindelöf property, and conversely, if R is connected, the former implies the latter. Hence a theorem that a regular space with the star-finite property is fully normal may be regarded as a generalization of the well-known Tychonoff's theorem.¹⁾ I believe that the star-finite property is more useful than the Lindelöf property in some cases.²⁾

In §1 we shall establish a few fundamental theorems concerning star-finite coverings and then these theorems will be applied in §§ 2 and 3. §3 is devoted to the proofs of the facts described above. Finally, in §4 some theorems will be given concerning locally bicompact spaces.

§ 1. Star-finite coverings.

Let $\mathcal{U}$ be an open covering of a T_1 -space R . For a subset A of R we denote by $S(A, \mathcal{U})$ the sum of all the sets of $\mathcal{U}$ which intersect A . If the covering $\{S(p, \mathcal{U}); p \in R\}$ (p ranging over all points of R) is a refinement of another covering $\mathcal{V}$, we shall say, after J. W. Tukey,³⁾ that $\mathcal{U}$ is a $\mathcal{A}$ -refinement of $\mathcal{V}$.

Theorem 1. Let $\mathcal{G} = \{G_\alpha; \alpha \in \Omega\}$ be a star-finite open covering of a T_1 -space R . Then $\mathcal{G}$ admits a star-finite open $\mathcal{A}$ -refinement $\mathcal{G}'$ such that the union of $\mathcal{G}$ and $\mathcal{G}'$ is also star-finite, if and only if there exists a closed covering $\mathfrak{F} = \{F_\alpha; \alpha \in \Omega\}$ such that $F_\alpha \subset G_\alpha, \alpha \in \Omega$.

Theorem 1 follows immediately from the following lemmas 1 and 2.

Lemma 1. Let $\mathcal{G} = \{G_\alpha; \alpha \in \Omega\}$ be a star-finite open covering of a T_1 -space R , and $\mathfrak{F} = \{F_\alpha; \alpha \in \Omega\}$ a closed covering of R such that $F_\alpha \subset G_\alpha, \alpha \in \Omega$. If we denote by $\mathcal{G}'$ the intersection of all the binary coverings

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1) A. Tychonoff, Ueber einen Metrisationssatz von P. Urysohn, Math. Ann. 95 (1925), pp. 139-142.

2) In particular, the star-finite property finds its applications in dimension theory. Cf. K. Morita, On the dimension theory of normal spaces, to appear in Jap. Jour. of Math.

3) J. W. Tukey, Convergence and uniformity in topology, Princeton, 1940. This book will be cited as T. in the following.

$$\{G_\alpha, R - F_\alpha\}, \alpha \in \Omega,$$

then the covering $\mathcal{G}'$ is an open Δ -refinement of $\mathcal{G}$ and the union of $\mathcal{G}$ and $\mathcal{G}'$ is star-finite.

Proof. A non-empty set G' of $\mathcal{G}'$ is written in the form

$$(1) \quad G' = \prod_{\alpha \in I} G_\alpha \cdot \prod_{\beta \in I'} (R - F_\beta)$$

where I is a subset of Ω . Since $\mathcal{G}$ is star-finite, I is a finite set. I' is not empty, for $\mathcal{G}$ is a covering of R . If we denote by I'' the set of all indices β such that $(\prod_{\alpha \in I} G_\alpha) \cdot G_\beta \neq 0$, then I'' is also a finite set. If $\beta \in I''$, then $(\prod_{\alpha \in I} G_\alpha) \cdot F_\beta = 0$.

Hence we have

$$\prod_{\alpha \in I} G_\alpha \leq R - F_\beta, \text{ for } \beta \in I'',$$

and consequently

$$G' = \prod_{\alpha \in I} G_\alpha \cdot \prod_{\beta \in I' - I''} (R - F_\beta).$$

Since I and I'' are finite sets, this shows that G' is an open set.

Let G be a set of $\mathcal{G}$ and G' a set of $\mathcal{G}'$. We shall write G' in the form (1), and denote by Δ the set of all indices α such that $G \cdot G_\alpha \neq 0$, $G_\alpha \in \mathcal{G}$. If $G \cdot G' \neq 0$, then we have $\Delta \cap \Delta' \neq \emptyset$. Since Δ is a finite set, the number of the sets G' of $\mathcal{G}'$ which intersect G is also finite. Hence the union of $\mathcal{G}$ and $\mathcal{G}'$ is star-finite, since $\mathcal{G}'$ is clearly a refinement of $\mathcal{G}$.

It remains to be proved that $\mathcal{G}'$ is a Δ -refinement of $\mathcal{G}$. Since $\mathcal{G}'$ is a refinement of the binary covering $\{G_\alpha, R - F_\alpha\}$, a set of $\mathcal{G}'$ is contained either in G_α or in $R - F_\alpha$. Hence a set G' of $\mathcal{G}'$ intersecting F_α must be contained in G_α , and consequently we have

$$S(F_\alpha, \mathcal{G}') \leq G_\alpha, \alpha \in \Omega.$$

$\{F_\alpha; \alpha \in \Omega\}$ being a covering of R there exists for each point p of R some F_α such that $p \in F_\alpha$. Therefore the covering $\mathcal{G}'$ is a Δ -refinement of $\mathcal{G}$.

Lemma 2. *If an open covering $\mathcal{G}'$ is a Δ -refinement of another open covering $\mathcal{G} = \{G_\alpha; \alpha \in \Omega\}$, then the closed sets*

$$F_\alpha = R - S(R - G_\alpha, \mathcal{G}'), \alpha \in \Omega$$

form a closed covering of R such that $F_\alpha \leq G_\alpha, \alpha \in \Omega$.

Proof. If a point p does not belong to any F_α , then we have $p \in S(R - G_\alpha, \mathcal{G}')$ for all $\alpha \in \Omega$. This shows that $S(p, \mathcal{G}') \cdot (R - G_\alpha) \neq 0$ for all α , which contradicts with the assumption of the lemma.

Theorem 2. Let $\mathcal{G} = \{G_\alpha; \alpha \in \Omega\}$ be a star-finite open covering of a T_1 -space R . Let us divide Ω into the classes F_c ($c \in C$) such that two indices c, β belong to the same class if and only if there exists a positive integer n for which $G_\alpha < S^n(G_\beta, \mathcal{G})$ holds, and denote by R_c the sum of all the sets G_α with $\alpha \in F_c$. Then we have

$$R = \sum R_c; R_c \cdot R_{c'} = 0, \quad \text{for } c \neq c', c, c' \in C,$$

and the sets R_c are open as well as closed, and $\{G_\alpha; \alpha \in F_c\}$ is a countable open covering of the space R_c .

Proof is obvious.

Theorem 3. Let $\mathcal{G} = \{G_1, G_2, \dots\}$ be a countable open covering of a T_1 -space R and $\mathfrak{F} = \{F_1, F_2, \dots\}$ a closed covering of R such that $F_i < G_i$, $i=1, 2, \dots$. If there exists for each i a real-valued continuous function $f_i(x)$ defined in R such that

$$f_i(x) = \begin{cases} 0, & x \in F_i \\ 1, & x \in R - G_i \end{cases}$$

then $\mathcal{G}$ admits a countable, star-finite, open Δ -refinement.

Proof. By the assumption of the theorem we can construct the open sets G_i^n ($i=1, 2, \dots; n=i, i+1, \dots$) such that

$$(2) \quad F_i < G_i^n < G_i, \quad \overline{G_i^n} < G_i^{n+1}, \quad n=i, i+1, \dots$$

Then, if we put

$$X_n = \sum_{i=1}^n G_i^n,$$

we have

$$R = \sum_{n=1}^{\infty} X_n; \overline{X_n} < X_{n+1}, \quad n=1, 2, \dots$$

Let us put further

$$H_n = X_n - \overline{X_{n-1}}, \quad K_n = \overline{X_n} - X_{n-1}, \quad n=1, 2, \dots,$$

where $X_0 = X_{-1} = X_{-2} = 0$. Then H_n is open and K_n is closed. Moreover we have

$$(3) \quad K_n < H_{n+1}, \quad \sum_{n=1}^{\infty} K_n = R$$

$$(4) \quad H_n \cdot H_m = 0, \quad \text{for } |n-m| \geq 3.$$

Since $K_n = K_n \cdot \overline{G_1^n} + \dots + K_n \cdot \overline{G_n^n}$, the system of closed sets $\{K_n \cdot \overline{G_i^n}; i=1, 2, \dots, n; n=1, 2, \dots\}$ forms a closed covering of R , and hence $\{H_{n+1} \cdot$

$G_i^{n+1}; i=1, 2, \dots, n; n=1, 2, \dots\}$ is also an open covering (by (2), (3)). From (4) we see that the latter covering is star-finite. Hence; if we construct the intersection $\mathcal{U}$ of all the binary coverings

$$\{H_{n+1} \cdot G_i^{n+1}, R - K_n \cdot \bar{G}_i^n\}, i=1, 2, \dots, n; n=1, 2, \dots,$$

then $\mathcal{U}$ is, by Lemma 1, a star-finite, countable, open $\mathcal{A}$ -refinement of $\mathcal{G}$.

§ 2. Normal spaces and normal coverings.

In case an open covering $\mathcal{G}$ admits a sequence of open coverings $\{\mathcal{G}_n\}$ ($n=1, 2, \dots$) such that $\mathcal{G}_n$ is a $\mathcal{A}$ -refinement of $\mathcal{G}_{n-1}$ and $\mathcal{G}_1$ is a refinement of $\mathcal{G}$, we shall say, after J. W. Tukey,⁴⁾ that $\mathcal{G}$ is normal.

Theorem 4. *A T_1 -space R is normal if and only if every star-finite open covering of R is normal.*

Lemma 3. *Let $\mathcal{G} = \{G_1, G_2, \dots\}$ be a countable open covering of a normal space R such that each point of R is contained in only a finite number of sets of $\mathcal{G}$. Then there exists a closed covering $\{F_1, F_2, \dots\}$ such that $F_i \subset G_i, i=1, 2, \dots$.*

Proof of Lemma 3. From the relation $R - \sum_{i=2}^{\infty} G_i \subset G_1$, it follows that there exists an open set H_1 such that $R - \sum_{i=2}^{\infty} G_i \subset H_1, \bar{H}_1 \subset G_1$. Then $\{H_1, G_2, G_3, \dots\}$ is an open covering. By an inductive process we can construct open sets $H_1, H_2, \dots$ such that

$$R - (H_1 + \dots + H_{n-1} + \sum_{i=n+1}^{\infty} G_i) \subset H_n, \bar{H}_n \subset G_n.$$

Then $\{H_1, H_2, H_3, \dots\}$ forms an open covering of R , because, if we take, for each point p of R , an integer n so large that $p \in R - \sum_{i=n+1}^{\infty} G_i$, we have $p \in H_1 + \dots + H_n$. Hence $\{\bar{H}_1, \bar{H}_2, \dots\}$ is a closed covering satisfying the condition of the lemma.

Lemma 4. *Let $\mathcal{G} = \{G_\alpha; \alpha \in \Omega\}$ be a star-finite open covering of a normal space R . Then there exists a closed covering $\{F_\alpha; \alpha \in \Omega\}$ such that $F_\alpha \subset G_\alpha, \alpha \in \Omega$.*

Lemma 4 follows readily from Lemma 3 and Theorem 2. Hence every star-finite open covering of a normal space admits a star-finite $\mathcal{A}$ -refinement by Lemma 1. This proves the "only if" part of Theorem 4. Since the "if" part is obvious, the proof of Theorem 4 is completed.

4) Theorems 4 and 7 are due to J. W. Tukey, but there is a mistake in his proofs. Cf. T. p. 49. The theorem 2.5 in T. p. 44 does not hold in general (even for a covering consisting of three sets).

Theorem 5. A countable open covering $\mathcal{G} = \{G_1, G_2, \dots\}$ of a normal space R is normal if and only if there exists a closed covering $\{F_1, F_2, \dots\}$ such that $F_i \subset G_i$, $i=1, 2, \dots$.

Proof. The "only if" part is obvious from Lemma 2. The "if" part follows from Theorems 3 and 4 and the normality of R .

Corollary. A countable open covering $\mathcal{G}$ of a normal space R is normal, if each point of R is contained in only a finite number of sets of $\mathcal{G}$.

Theorem 6. A countable open covering of a normal space is normal, if and only if it admits a countable star-finite open refinement.

Proof. The "only if" part is proved by virtue of Lemma 2 and Theorem 3. The "if" part follows directly from Theorem 4.

Now we shall proceed to the case of normal coverings of a T_1 -space.

Theorem 7. If $\mathcal{G}$ is a star-finite normal (open) covering of a T_1 -space R , then $\mathcal{G}$ has a star-finite Δ -refinement $\mathcal{G}'$ such that $\mathcal{G}'$ is a normal covering and the union of $\mathcal{G}$ and $\mathcal{G}'$ is star-finite.

For the proof of Theorem 7, it is sufficient, in view of Lemmas 1 and 2, to prove the following two lemmas.

Lemma 5. If $\mathcal{G}$ is a normal covering of a T_1 -space R , then the binary covering $\{G, S(R-G, \mathcal{G})\}$ is normal for any open set G of R .

Lemma 6. (Supplement to Lemma 1) Besides the assumption of Lemma 1 let us assume further that the binary coverings $\{G_\alpha, R-F_\alpha\}$ ($\alpha \in \Omega$) are all normal. Then the covering $\mathcal{G}'$ defined there is also normal.

Proof of Lemma 5. According to the theorem 9.3 in T. p. 53, a binary open covering $\{U_1, U_2\}$ of a T_1 -space R is normal if and only if there exists a real-valued continuous function $f(x)$ such that

$$f(x) = \begin{cases} 0, & x \in R - U_1, \\ 1, & x \in R - U_2. \end{cases}$$

The existence of such a function for the covering $\{G, S(R-G, \mathcal{G})\}$ is assured by the theorem 9.2 in T. p. 53. Hence we have Lemma 5.

Lemma 6 is proved by virtue of the lemma 5.2 in T. p. 49, since

$$G_\alpha \circ \{G_\alpha, R-F_\alpha\} = \{G_\alpha, R-F_\alpha\}.$$

Theorem 3 and Lemmas 2,5 lead us to the following theorem (Cf. also the beginning of the proof of Lemma 5).

Theorem 8. If a countable open covering $\mathcal{G}$ of a T_1 -space R is normal, then $\mathcal{G}$ admits a countable star-finite open Δ -refinement.

§ 3. The star-finite property.

A T_1 -space R is called fully normal, if every open covering of R is normal, or every open covering admits a $\mathcal{A}$ -refinement. (J. W. Tukey). A fully normal space is always normal.

Theorem 9. *A regular space with the star-finite property is fully normal.*

Proof. We shall first prove that R is normal. Let A and B be disjoint closed sets of R . For each point p of A we construct a neighbourhood $U(p)$ such that $\overline{U(p)} \cdot B = 0$. For a point p of B we construct $U(p)$ so that $\overline{U(p)} \cdot A = 0$. Then these neighbourhoods $U(p)$ ($p \in A+B$) and the set $R-(A+B)$ constitute an open covering $\mathcal{U}$ of R . From the star-finite property there exists a star-finite open covering $\mathcal{G}$ which is a refinement of $\mathcal{U}$. By Theorem 2 we see that by means of the covering $\mathcal{G}$ the space R decomposes into the sum of mutually disjoint sets R_c ($c \in C$) each of which is closed and open simultaneously, and the sets of $\mathcal{G}$ intersecting R_c form a countable covering of R_c .

Let us put

$$A_c = A \cdot R_c, \quad B_c = B \cdot R_c$$

and consider A_c, B_c in the space R_c . If we denote the sets of $\mathcal{G}$ intersecting A_c by $X_1, X_2, \dots$ and the sets of $\mathcal{G}$ intersecting B_c by $Y_1, Y_2, \dots$, then we have

$$\begin{aligned} A_c &< X_1 + X_2 + \dots, & \overline{X_n} \cdot B &= 0 \\ B_c &< Y_1 + Y_2 + \dots, & \overline{Y_n} \cdot A &= 0 \end{aligned}$$

If we put

$$\begin{aligned} U_1 &= X_1, & V_1 &= Y_1 - \overline{X_1}, \\ U_n &= X_n - (\overline{V_1} + \dots + \overline{V_{n-1}}), & V_n &= V_n - (\overline{U_1} + \dots + \overline{U_{n-1}}), \quad n=2, 3, \dots \end{aligned}$$

and

$$G_c = \sum_{n=1}^{\infty} U_n, \quad H_c = \sum_{n=1}^{\infty} V_n,$$

we have

$$A_c < G_c < R_c, \quad B_c < H_c < R_c, \quad G_c \cdot H_c = 0^{4a)}$$

Hence, if we put

$$G = \sum_{c \in C} G_c, \quad H = \sum_{c \in C} H_c,$$

we have

$$A < G, \quad B < H, \quad G \cdot H = 0.$$

Since G and H are clearly open, this proves that R is normal.

Now that the normality of R is proved, Theorem 9 follows readily from the star-finite property and Theorem 4.

4a) Cf. A. Tychonoff, loc. cit.

Theorem 10. *A regular space R with the Lindelöf property (i.e., every open covering of R is reducible to a countable covering) possesses the star-finite property.*

Proof. Let $\mathcal{G}$ be an open covering of R . We shall show that $\mathcal{G}$ admits a star-finite open refinement. For this purpose we proceed as follows. Let us construct for each point p of R a neighbourhood $U(p)$ such that $\overline{U(p)}$ is contained in some set of $\mathcal{G}$. By the Lindelöf property we can find a countable number of points $p_1, p_2, \dots$ such that $\sum_{i=1}^{\infty} U(p_i) = R$. From the construction of $U(p)$ there exists for each point p_i a set G_i of $\mathcal{G}$ such that $\overline{U(p_i)} < G_i$.

Since R is normal by Tychonoff's theorem (Cf. the proof of Theorem 9), we can apply Theorem 3 to the coverings $\{G_i; i=1, 2, \dots\}$ and $\{\overline{U(p_i)}; i=1, 2, \dots\}$. Thus $\mathcal{G}$ has a star-finite refinement. This completes the proof.

As an immediate consequence of Theorem 10 and 2 we have

Corollary. *For a connected regular space the star-finite property is equivalent to the Lindelöf property.*

Remark. There is a space which has the star-finite property, but not the Lindelöf property. A fully normal space does not necessarily possess the star-finite property, since a connected non-separable metric space has not the star-finite property by the above corollary. However, it is to be noted that every countable open covering of a fully normal space admits a countable star-finite refinement, that is, a fully normal space has, so to speak, the star-finite property in the weak sense (Cf. Theorem 6).

Theorem 11. *Let R be a regular space. If there exists an open covering $\mathcal{G}$ of R such that each set of $\mathcal{G}$ has the Lindelöf property and $\mathcal{G}$ admits an open star-refinement $\mathcal{H}$ (i.e., for each set H of $\mathcal{H}$, $S(H, \mathcal{H})$ is contained in some set of $\mathcal{G}$),⁵⁾ then R decomposes into the sum of mutually disjoint open sets each of which has the Lindelöf property, and R is a fully normal space with the star-finite property.*

Proof. In case there exists an integer n such that $y \in S^n(x, \mathcal{H})$ we shall write $x \sim y$ (x, y being points of R). By this relation we can decompose R into the sum of mutually disjoint sets R_α ($\alpha \in \mathcal{Q}$) such that x, y belong to the same set R_α if and only if $x \sim y$.

Then R_α is the sum of the sets $S^n(x, \mathcal{H})$, $n=1, 2, \dots$ for each point x of R_α . Since $R_\alpha \cdot R_\beta = \emptyset$ for $\alpha \neq \beta$, R_α is both open and closed.

Now we shall prove that R_α has the Lindelöf property. For the sake of brevity we shall say that a subset A of R has the Lindelöf property *with respect to R* , if from any open covering $\mathcal{U}$ of R we can always select a countable number

5) T. pp. 44-45.

of sets $M_1, M_2, \dots$ of $\mathcal{G}$ such that $A \subset M_1 + M_2 + \dots$. Then, if A has the Lindelöf property with respect to R , $S(A, \mathcal{G})$ has also the same property. Because, by the assumption, there exist a countable number of sets $H_1, H_2, \dots$ of $\mathcal{G}$ such that $A \subset H_1 + H_2 + \dots$; hence we have $S(A, \mathcal{G}) \subset S(H_1, \mathcal{G}) + S(H_2, \mathcal{G}) + \dots \subset G_1 + G_2 + \dots$, where G_i are suitable sets of $\mathcal{G}$ and consequently $S(A, \mathcal{G})$ has the Lindelöf property with respect to R . Therefore $S^n(x, \mathcal{G})$ has the Lindelöf property with respect to R for each n , and hence R_α has also the same property. Since R_α is a closed set of R , R_α has the Lindelöf property in the usual sense.

Thus the first part of Theorem 11 is proved. The second part follows readily from Theorem 9 and 10. Therefore the proof of the theorem is completed.

Corollary 1. *In the hypothesis and the conclusion of Theorem 11 we can replace "has the Lindelöf property" by "is perfectly separable".*

Since metrizable spaces are fully normal⁶⁾, we have at once

Corollary 2. *If a metrizable space R is locally separable, then R decomposes into the sum of mutually disjoint open sets each of which is separable.⁷⁾*

§ 4. Locally bicom pact spaces.

Theorem 12. *If, for a Hausdorff space R , there exists an open covering $\mathcal{G}$ such that the closure of each set of $\mathcal{G}$ is bicom pact and $\mathcal{G}$ admits a star-refinement $\mathcal{H}$, then R is a fully normal space with the star-finite property.*

Proof. Any set of $\mathcal{G}$ has the Lindelöf property with respect to R and in the proof of Theorem 11 we have used only the fact that each set of $\mathcal{G}$ has the Lindelöf property with respect to R . Hence we have Theorem 12 by Theorem 11.

A locally bicom pact Hausdorff space is completely regular, but not necessarily normal. In this connection the following theorem is of some interest.

Theorem 13. *In order that a locally bicom pact Hausdorff space R be fully normal it is necessary and sufficient that R possess the star-finite property.*

This theorem follows from Theorems 9 and 12.

Theorem 14. *A locally bicom pact, fully normal space is uniformly locally bicom pact (with respect to some uniform structure) in the sense of A. Weil⁸⁾, and conversely.*

Proof. Let R be a locally bicom pact, fully normal space, and $\mathcal{G}$ an open covering of R such that the closure of each set of $\mathcal{G}$ is bicom pact. If $\mathcal{H}$ is a $\star$ -refinement of $\mathcal{G}$, then the closure of $S(p, \mathcal{H})$ is bicom pact for every point p

6) T. p. 53.

7) This is a theorem due to P. Alexandroff and W. Sierpinski. Cf. P. Alexandroff, Über die Metrisation der im Kleinen bikompakten topologischen Räume, Math. Ann., **92** (1924), pp. 294-301. W. Sierpinski, Sur les espaces métriques localement séparables, Fund. Math., **21** (1933), pp. 107-113.

of R . This proves the first part of the theorem. The other part follows readily from Theorem 12.

Corollary. *A locally bicomact metrizable space R admits a metric $\rho(x, y)$ ($x, y \in R$) such that the closure of the ε -neighbourhood $U(x; \varepsilon)$ of a point x with respect to the metric ρ is bicomact for every point x and for some positive number ε (independent of x).*

Proof. Let ρ_0 be a metric of R and $\mathcal{G}_n$ an open covering of R which consists of 2^{-n} -neighbourhoods $U_0(x; 2^{-n})$ of x with respect to ρ_0 (x ranging over all points of R). Let $\mathcal{W}_1$ be an open covering such that the closure of each set of $\mathcal{W}_1$ is bicomact. Then we construct successively open coverings $\mathcal{W}_2, \mathcal{W}_3, \dots$ such that $\mathcal{W}_n$ is a star-refinement of $\mathcal{W}_{n-1}$ as well as a refinement of $\mathcal{G}_{n-1}$. This is possible, since metric spaces are fully normal.⁸⁾ From the sequence $\{\mathcal{W}_1, \mathcal{W}_2, \dots\}$ we can construct, by a well-known process,⁹⁾ a new metric ρ of R such that $\rho(x, y) < 2^{-n-1}$ implies $y \in S(x, \mathcal{W}_n)$. The metric ρ thus obtained clearly satisfies the condition of the corollary.

We shall remark here that a locally bicomact topological group, when considered as a space, is a fully normal space with the star-finite property.

Theorem 15. *In order that a fully normal space be locally bicomact and of dimension at most n it is necessary and sufficient that there exist a system of star-finite open coverings of order at most $n+1$ such that the union of each pair of these coverings be star-finite and every open covering admit some covering of this system as a refinement.*

Proof. The necessity. Let $I = \{\mathcal{W}_\alpha; \alpha \in \Omega\}$ be the totality of all star-finite open coverings $\mathcal{W}_\alpha$ of order at most $n+1$ such that the closure of each set of $\mathcal{W}_\alpha$ is bicomact. Then it is easily shown, by a theorem proved in our previous paper¹⁰⁾, that I satisfies the conditions of the theorem.

The sufficiency. Let $I = \{\mathcal{W}_\alpha; \alpha \in \Omega\}$ be a system of open coverings $\mathcal{W}_\alpha$ with the mentioned property. Then it follows easily that the dimension of R is at most n and the closure of each set of $\mathcal{W}_\alpha$ is bicomact.

Remark. On the basis of this theorem a new homology theory of locally bicomact fully normal spaces may be developed.

8) A. Weil, Sur les espaces à structure uniforme et sur la topologie générale, Actualités Sci. Ind. No. 551, Paris 1937, p. 27. In pp. 35-36 of this book A. Weil says, "Il semble que sur un espace connexe, localement compact, dénombrable à l'infini, la class de tous les recouvrements ouverts compacts localement finis soit régulière". This proposition follows readily from our theorems.

9) A. Weil, loc. cit., p. 14; or T. p. 51.

10) K. Morita, loc. cit., 2).

On Power Series in Abstract Spaces.

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Many theorems of analytic functions has been extended to the complex Banach spaces by A. E. Taylor¹⁾ and others. It is our purpose here to investigate some characteristics of a power series in the complex Banach spaces. Let E and E' be two complex Banach spaces. An E' -valued function $f(x)$ defined on a domain D of E is analytic in D if it is strongly continuous on D and if it admits a Gateauw differential, i.e. G -differentiable, at every point of D . The following two theorems²⁾ will have to be used for our purpose.

Theorem A. Let $p(x)$ satisfy the following conditions:

- 1). it is G -differentiable on E ,
- 2). for $|\xi|=1$, $\|P(\xi x)\| = \|P(x)\|$,
- 3). there exists an x_0 in E and real number M, σ , with $\sigma > 0$, such that for $\|x - x_0\| \leq \sigma$ we have $\|P(x)\| \leq M$.

Then $\|P(x)\| \leq M$ for $\|x\| \leq \sigma$.

Theorem B. If an E' -valued function $f(x)$ is G -differentiable and B -continuous on an open set D of E it is analytic on D .

The series of the form $f(x) = \sum_{n=0}^{\infty} h_n(x)$ is called "Power series", where $h_n(x)$ is a homogeneous polynomial of degree n .³⁾

Definition I. If a positive number λ satisfies the following conditions: 1) if $0 < r < \lambda$, $f(x)$ is analytic and bounded in the sphere defined by $\|x\| < r$, 2) if $r > \lambda$, $f(x)$ can not be analytic and bounded in the sphere defined by $\|x\| < r$, λ is called a radius of bound of $f(x)$.

Definition II. We call a positive number τ a radius of analyticity of $f(x)$, when the following conditions are satisfied, 1) $f(x)$ is analytic in the sphere defined by $\|x\| < \tau$, 2) $f(x)$ is not analytic in the sphere defined by $\|x\| < r$, where $r > \tau$.

1). A. E. Taylor (1). Analytic functions in general analysis. Annali della R. Scuola Normale Superiore di Pisa, Seri II, Vol. VI (1937). (2): On the properties of analytic functions in abstract spaces, Math. Ann. 115, 193.

2). Max. A. Zorn: Characterization of analytic functions in Banach spaces, Annals of Math. Vol. 45 (1945).

3). See A. E. Taylor (2), page 467, Definition 3.